When Senses Collide: Investigating Modality Congruence and Interference Between Task and Notification in Augmented Reality

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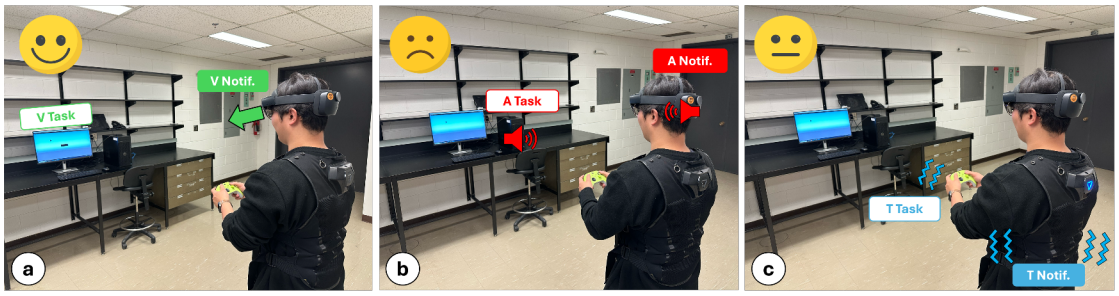


Figure 1: Examples of crossmodal multitasking scenarios used in our study: (a) Visual notification (3D arrow) on an AR display during a visual pattern recognition task presented on a monitor, (b) Spatialized audio notification from the AR display during an auditory pattern recognition task played on a workstation PC, and (c) Directional tactile notification via a wearable vest during a tactile vibration pattern recognition task using a handheld controller.

Abstract

As augmented reality (AR) technologies become more integrated into everyday tasks, the design of notification systems that minimize disruption while enhancing user awareness is increasingly important. This study investigates how crossmodal interference between notification modality (visual, audio, tactile) and primary task modality (also visual, auditory, tactile) affects user perception and performance in AR environments. Through a controlled user study ($N = 36$), participants engaged in modality-specific pattern recognition tasks while responding to spatial directional notifications delivered via different sensory channels. By analyzing notification awareness time, reaction time, task accuracy, and subjective measures, such as cognitive load and user preference, the study reveals how congruency or mismatch between task and notification modalities can either facilitate or hinder attention redirection and multitasking efficiency. The findings contribute to the design of adaptive AR notification systems that are less intrusive and better aligned with users' perceptual and cognitive states.

Index Terms: Directional notification, crossmodal interference, augmented reality, user perception, task performance.

1 Introduction

As daily life becomes increasingly complex, individuals increasingly rely on technology to manage, track, and navigate their tasks [45]. Multitasking, in particular, has become a central aspect of our modern routines, where individuals often engage in multiple concurrent tasks that demand varying levels of attention and cognitive load. Digital notification systems play a crucial role in keeping users informed, directing their attention, and prompting timely actions, making them indispensable for managing multiple tasks [35].

Over the past decades, research on notification systems—particularly in mobile and wearable computing—has investigated how various sensory modalities (e.g., visual, auditory, tactile) can be leveraged to deliver alerts with minimal disruption [14]. These studies have informed the design of more effective and less intrusive notifications across a range of contexts [28]. However, with the emergence of immersive platforms like virtual and augmented reality (VR/AR), notification design is being recontextualized within perceptually richer and spatially embedded environments.

Among these platforms, AR offers unique opportunities to deliver directional, context-sensitive notifications directly into the user's perceptual space [22]. In such settings, users must remain situationally aware of both their physical and virtual surroundings, making the timing, modality, and spatial positioning of notifications especially critical [31]. The use of *directional notifications* in AR is particularly promising as cues not only indicate the occurrence of an event but also guide users toward a specific spatial location in the augmented scene. These notifications can be implemented through visually (e.g., arrows), auditorily (e.g., spatialized sounds), or tactilely (e.g., vibration cues via wearable actuators) [3, 6]. Despite the growing interest in these techniques, little is known about how such notifications interact with ongoing user tasks that engage specific sensory modalities [54].

Understanding these dynamics is essential for designing attention-aware AR interfaces. Modality *congruence*—when the notification shares the same sensory channel as the primary task—may either help by reinforcing perceptual salience or hinder by overloading that channel. Conversely, modality *incongruence*—using different channels—may reduce perceptual interference or introduce split attention challenges. Clarifying these trade-offs is especially important for designing adaptive and non-intrusive notification systems in AR. Most of existing work has mostly evaluated notification modalities in isolation or in dual-mode scenarios (e.g., visual-auditory). Few studies have systematically examined cross-

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modal interactions—that is, how the sensory modality of the notification interacts with that of the task being performed. However, it remains unclear whether a tactile notification is more effective when the user is visually or tactilely engaged, or whether such a modality congruence might instead introduce cognitive interference due to sensory overload.

To address this gap, we conducted a 3×3 within-subject user study that systematically varied the task modality (visual, auditory, tactile) and the notification modality (visual, auditory, tactile). Participants performed modality-specific pattern recognition tasks while intermittently receiving directional notifications that prompted them to locate a spatially indicated virtual object in AR. This study design allowed us to explore not only the individual effectiveness of each notification modality but also the interaction effects between task and notification modalities on user performance, cognitive load, and subjective preference.

In this study, we pose the following research questions (RQs):

RQ1 How does the modality of a directional notification (visual, auditory, or tactile) affect user performance and perception in AR environments, independently of the task modality?

RQ2 How does the combinations of task and notification modalities (congruence or incongruence) impact task accuracy, preference, and perceived task load and ease of use?

By addressing these questions, we aim to inform the design of modality-aware notification strategies for AR systems that enhance multitasking efficiency while minimizing cognitive disruption.

2 Related Work

Previous research has explored effective notifications with the goal of designing less disruptive and user-centered notification delivery using mobile [16] and wearable devices [47]. This section reviews related works within the context of immersive environments.

2.1 Notification Modalities in Augmented Reality

Various modalities have been explored for notification delivery, including visual, audio, and haptic feedback [42]. Visual feedback conveys information through visual elements like icons, text, or animations [48]. Audio feedback uses sound cues such as alerts or instructions [4]. Haptic feedback communicates through tactile sensations like vibrations or force [20]. Previous studies have investigated the effectiveness of these modalities for delivering notifications in AR. For instance, Plabst et al. [38] examined spatial placement strategies for visual notifications, including display-fixed, body-fixed, and world-fixed coordinates. Cidota et al. [8] compared visual and audio notifications in a collaborative AR setting and found that visual cues were generally preferred over auditory ones to maintain task awareness.

Earlier studies have also highlighted the potential benefits of using multimodal notifications over unimodal ones. Lazaro et al. [29] compared unimodal visual and auditory notifications with multimodal notifications in AR and found that multimodal cues achieved the highest recognition rates. This suggests that sensory redundancy—the simultaneous presentation of the same information across multiple sensory modalities—enhances the noticeability of notifications. Similarly, Raikwar et al. [41] showed that combining spatialized audio with object-fixed visual cues improved performance and satisfaction in an AR-based cooking task.

While these studies provide valuable insights into modality choices and interface design in immersive environments, notification effectiveness has primarily been evaluated in isolation from the nature of users' primary tasks. To address this gap, this work systematically investigates the interplay between primary task types and notification modality.

2.2 Directional and Spatial Notification Cues

Another line of research has emphasized the importance of spatial and directional cues in notification systems within immersive environments [1]. Spatial awareness enhances user experience by promoting presence and enabling intuitive engagement with physical and virtual content. Delivering spatially-embedded notifications in immersive environments is crucial for redirecting users' attention without disrupting their sense of immersion [36, 52]. Stefanucci [51] demonstrated that AR-style navigational cues in VR simulations can enhance spatial knowledge acquisition without increasing cognitive load.

Earlier research has leveraged visual, auditory, and haptic cues to provide directional and spatial notifications. Zhang et al. [59] introduced 3D spatial visual cues in MR remote collaboration, demonstrating improved efficiency and spatial awareness in complex assembly tasks. Raikwar et al. [41] employed directional spatial audio notifications, and Lordianto et al. [30] proposed a construction helmet with directional haptic feedback to enhance safety through spatially-oriented tactile alerts. Gibson et al. [18] found visual-tactile navigation aids enhance obstacle avoidance, albeit with increased cognitive load from tactile feedback. However, these studies neglect how spatial notifications interact with concurrent task modalities and their cognitive demands - a gap our work addresses.

2.3 Notification Modalities in Multitasking

Users often deal with notifications while they are doing specific tasks. To effectively deliver notifications, it is essential to understand the interaction between task modality and notification modality to prevent sensory overload, reduce task interference, and facilitate timely recognition of notifications. Notification acceptance refers to the degree of disruption users experience due to notifications, their behavioral responses, and their subjective perceptions [2, 15]. These challenges can be understood through the lens of Wickens' Multiple Resource Theory (MRT) [57], a framework for how humans process information across different sensory channels and cognitive stages. It suggests that using different modalities for task and notification, such as separating sensory (e.g., auditory vs. visual) and cognitive (e.g., spatial vs. temporal) channels, can help reduce interference and improve performance. However, there is a lack of a standardized notification management system [17]. Enhancing this acceptance requires understanding how users perceive and respond to different modalities within the context of their tasks [26].

Previous research highlights that modality effectiveness depends on task context. For example, Ghosh et al. [17] highlighted that users often miss physical, digital, or temporal events while immersed in VR. With modality effectiveness varying by event type: visual-tactile combinations worked best for physical events, while auditory cues were preferred for digital ones; overall, multimodal notifications outperformed unimodal ones in speed and accuracy, supporting MRT's predictions [57].

Recent AR research has explored optimal notification modalities during real-world tasks. Marques et al. [32] first established tactile notifications as most effective for capturing attention in asynchronous MR collaboration scenarios. A follow-up study [33] confirmed this across both on-site and remote users. However, the question of how the alignment (congruency) or misalignment (incongruency) between the user's task modality (e.g., visual, tactile, auditory) and the notification modality affects user performance has not been clearly explored. For example, it remains unclear how user performance differs when a visual notification is delivered during a visually demanding task versus when a tactile or auditory notification is presented.

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3 Materials: Modality-Specific Tasks and AR-Based Directional Notifications

3.1 Crossmodality Multitasking Scenario

For our study, we designed crossmodality multitasking scenarios involving two concurrent tasks: a primary task and a secondary task. The primary task required modality-specific pattern recognition, while the secondary task involved responding to AR-based directional notifications. We developed nine modality conditions, consisting of combinations of three task modalities (V_t: visual, A_t: auditory, T_t: tactile) and three notification modalities (V_n: visual, A_n: auditory, T_n: tactile).

During trials, participants continuously engaged in their assigned modality-specific task—identifying Morse-like sequences and responding via handheld controller by pressing "X" for correct patterns or "Y" for incorrect ones (mapping to gaming console conventions). Simultaneously, they received randomly timed crossmodal notifications (visual arrows, spatial audio pings, or controller vibrations) that redirected attention to AR spatial targets. Participants were required to balance these competing demands, maintaining pattern recognition accuracy in the primary task while locating and selecting notification targets within a constrained 7-second response window. This dual-task paradigm intentionally introduced modality conflicts, particularly when notification and task modalities overlapped (e.g., visual notifications during visual pattern recognition). Task and notification details are described below.

3.2 Apparatus & Resources

We developed a Unity-based application (v2022.3.15f1) to support crossmodality multitasking, integrating all components for automated task and notification delivery. Our hardware comprised:

- **Display/Interaction**: Microsoft HoloLens 2 (for AR rendering and interaction), an AR HMD with a 50° field of view (FOV), a resolution of 2048×1080 for each eye with a $75 Hz$ refresh rate.
- **Input Interface**: Microsoft Xbox Controller (for primary task control, ensuring consistent input latency (approximately 15 to 20 ms)). This features dual-motor vibration feedback, to create different vibration effects, one larger motor and one smaller motor. It allows us to adjust the vibration intensity; we used the maximum setting, 1.0 on a scale from 0.0 to 1.0.
- **Haptic Feedback**: bHaptics X40 Tactsuit is a wearable haptic vest that features a total of 40 vibrotactile haptic feedback points (vibration motors) distributed across the vest. The suit was adjustable to fit all participants.

3.3 Task: Modality-Specific Pattern Recognition

The primary task requires participants to recognize structured sequences composed of short and long signals delivered through a single sensory channel, thereby isolating modality-specific cognitive demands. Our choice of using temporally structured short/long pattern recognition tasks is grounded in cognitive research that highlights the efficacy of such designs in assessing attentional allocation, working memory, and perceptual discrimination across sensory modalities [24, 49]. They offer a high degree of control over stimulus timing and complexity, which is critical for comparing performance across modalities in multitasking and attention research. The following details the development process, theoretical rationale, and specific design considerations.

Visual Task (V_t) The visual task requires users to identify Morse-code-like patterns composed of alternating shapes—a dot (short) and a dash (long) (see Figure 2-left). These patterns are visually presented in short sequences. Such visual pattern recognition tasks are widely employed to assess visual attention, cognitive load, and task-switching performance [27, 37]. In our study, each trial involves a brief presentation of a pattern sequence, followed by a short delay, during which participants must rely on their visual short-term memory to recall and identify the pattern from a set of options. Each trial presented a 4-element pattern sequence. Individual shapes appeared for 0.5 seconds followed by a blank screen for 0.5 seconds (1 second per element). The total duration for each pattern was 4 seconds. There was a gap of 3 seconds between patterns, to give the participants time to decide whether the pattern was correct or incorrect. The task presented a sequence of 20 patterns to the participants.



Figure 2: Conceptual illustrations of the task modalities.

Auditory Task (A_t) Participants were required to recognize auditory sequences composed of short and long-duration tones played on a workstation PC. Auditory pattern recognition tasks based on tone duration and pitch have commonly been utilized in previous research to assess auditory attention and cognitive workload [24, 56]. This task is analogous to the visual task, with short tones representing dots and longer tones representing dashes. Each pattern was 4 elements long (Figure 2-middle). The audio was a distinctive electronic beep that was played with a duration difference to separate the dot and the dash. The dot was a 0.3 second long electronic beep, followed by a 0.7 second gap. The dash was a 0.5 second long beep followed by a 0.5 second gap. The time intervals were validated as distinguishable but challenging by the internal members. This aligns with auditory discrimination thresholds showing 0.2 second as the minimal detectable duration difference for tones [21]. There was a 3-second long gap between patterns to give participants time to make a decision. Participants listened to each pattern and were then prompted to identify the sequence. The duration of the task was matched to the visual trials to enable direct crossmodal comparison, while the simple electronic beeps ensured that the task focused on temporal pattern recognition without unnecessary spectral complexity that might interfere with spatial audio notifications. The volume was at 100% to ensure.

Tactile Task (T_t) Similarly, the tactile primary task involved participants identifying vibration patterns using an Xbox handheld controller (Figure 2-right). Following the same structure as the visual and auditory tasks, short 0.3 second vibrations represented dots (followed by 0.7 second pauses) while 0.5 second vibrations represented dashes (followed by 0.5 second pauses), creating standardized 4-second pattern sequences. Participants received twenty 4-element patterns through the controller (at 100% vibration intensity) and the patterns had the same 3 second interval to let the user decide on an answer. The use of distinct tactile patterns is well-established in studies examining tactile attention, cognitive load, and interface usability [6]. This implementation maintained direct comparability with our other sensory tasks through identical temporal structures and difficulty levels while ensuring clear perceptibility through distinct yet comfortable vibration cues.

3.4 Notification: AR-Based Direction Recognition

The secondary task required participants to respond to directional notifications presented intermittently during the execution of each

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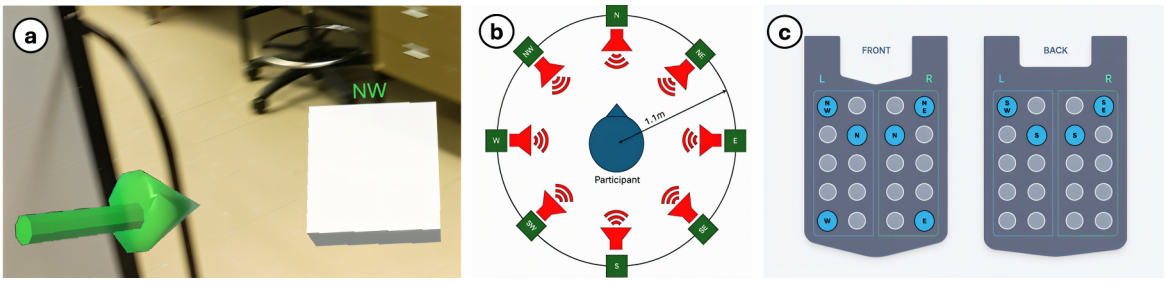


Figure 3: Directional notification modalities: (a) Visual (3D green arrow for indicating directions), (b) Auditory (audio sources depending on directions), and (c) Tactile (vibration positions mapped with directions)

primary task. Upon detecting a notification, participants were instructed to momentarily pause the ongoing primary task and shift their attention to select a specified virtual object within the AR environment (a virtual box in Figure 3a).

The purpose of the secondary task was to introduce a layer of spatial attention and sensorimotor coordination, encouraging participants to perform quick perceptual-motor actions while maintaining cognitive performance on the primary task. This simulates a scenario in which individuals must monitor for environmental cues while engaged in a focal task, such as responding to system alerts during equipment monitoring or navigating while processing spatial instructions. The following details the development of the AR directional notifications along with their theoretical rationale.

Visual Notification (V_n) A 3D virtual arrow was used as the visual notification modality (Figure 3a). The arrow was designed with a high-contrast color scheme to enhance visibility against varying backgrounds, while its peripheral placement was intended to balance saliency with minimal cognitive disruption. To minimize interference with the primary task, the arrow was displayed in the lower-left corner of the participant's field of view, ensuring it remained visible without obstructing primary task elements. Upon activation, the arrow dynamically appeared and persistently pointed to the location of the target virtual box for a duration of 5 seconds, providing participants with a clear and consistent directional cue. This design aligns with AR practices, using off-center cues to guide attention without overloading the primary visual field [3].

Audio Notification (A_n) Spatialized audio cues were designed and implemented using Unity's built-in spatial audio features, providing directional sound localization cues via the HoloLens 2's integrated headphones. Each audio cue was emitted from the exact spatial location of the target object, guiding the participant's attention through auditory spatial perception (see Figure 3b). To make the notifications distinct from the background noise and the auditory task, we used short high-pitched pings that were repeated every 0.5 seconds. These pings lasted 5 seconds, matching the duration of the visual notifications. The volume was kept at 80% to make it noticeable and to avoid startle reactions. The spatialized tones dynamically adjusted based on the participant's head orientation, reinforcing the perception that the sound originated from the target's actual position in space. This approach utilized existing evidence that spatial audio effectively supports directional attention and navigation tasks in AR [55, 25].

Tactile Notification (T_n) A wearable haptic vest with multiple embedded vibration actuators was employed to deliver directional tactile feedback. These actuators were spatially distributed along the sides of the vest to correspond with the left and right directions relative to the user's centered position in AR (see Figure 3c). Vibrations were triggered via Bluetooth communication from the Unity AR application, activating the actuators located on the side corresponding to the direction of the virtual box. Similar to the other notifications, the actuators vibrated for 5 seconds. To ensure the vibration was noticeable across different clothing types (e.g. thicker sweatshirts or thin shirts), we set the vibration intensity to their maximum strength. There were different actuators corresponding to each of the 8 virtual box directions (Figure 3c). This method was inspired by earlier research demonstrating wearable tactile feedback's effectiveness for spatial and directional cues [54].

4 Method: A Controlled User Study

4.1 Study Design

This section presents our study design to investigate the effects of crossmodal (mis)alignment between task and notification modalities on user perception performance in AR. It employes a within-subjects study consisting of two independent factors: **Task Modality** (Visual, Auditory, Tactile) and **Notification Modality** (Visual, Audio, Tactile), which are described in Section 3. This 3×3 factorial design allows us to examine both matched and mismatched conditions across the nine possible combinations while minimizing variability due to individual differences among participants.

Task Modality We incorporated three sensory modalities for the primary pattern recognition tasks (see Figures 1 and 2). In each task, participants were first presented with a target pattern and were then asked to identify the same pattern from a set of alternatives.

- **Visual Task** (V_t): Sequences of two alternating shapes (short(dot)/long(dash)) on screen.
- **Auditory Task** (A_t): Sequences of spatial audio tones with varying duration (short/long) through an AR headset.
- **Tactile Task** (T_t): Sequences of vibration patterns (short/long) through a hand-held controller.

Notification Modality During the primary task, participants were intermittently interrupted by directional notifications indicating the location of a virtual box within the AR space. In each trial, 5 such notifications were presented, each appearing at a random direction associated with a specific sensory modality. These notifications were delivered through one of three modalities (Figure 3):

- **Visual Notification** (V_n): A 3D arrow appeared in the AR view, pointing to the virtual object.
- **Audio Notification** (A_n): A spatial audio cue played from the direction of the virtual object.
- **Tactile Notification** (T_n): A vibrotactile feedback delivered through a wearable haptic vest on the side corresponding to the object's location.

4.2 Hypotheses

Wickens' MRT [57] discussed that notifications delivered through a modality that is different from that of the primary task (i.e., crossmodal approach) are expected to cause less interference. It also described visual channels are more efficient for spatial and directional information. Accordingly, we established the following hypotheses:

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- **H1**: Visual notifications (Vn) will demonstrate **(a)** higher task performance—including faster reaction times and greater spatial localization accuracy—compared to auditory (A_n) and tactile (T_n) notifications, and **(b)** higher subjective ratings in ease of use and preference than other modalities.
- **H2**: Auditory notifications (A_n), when the task modality is also auditory, will **(a)** reduce primary task accuracy and **(b)** increase perceived task load due to sensory and temporal interference.
- **H3**: Tactile notifications (T_n) will **(a)** be detected more quickly and accurately than auditory (A_n), but more slowly than visual (V_n), across all task conditions, and **(b)** be rated as easier to use and less mentally demanding than auditory (A_n), but less so than visual (V_n), across all task conditions.
- **H4**: Notification-task modality *mismatch* will **(a)** yield better notification response accuracy and **(b)** result in lower overall cognitive load than matched conditions.

These hypotheses are grounded in established theoretical frameworks and prior research findings. **H1** is supported by previous studies on AR interfaces, which demonstrate that visual cues offer superior performance for conveying directional information [3, 9]. **H2** is based on evidence of auditory channel overload effects, where overlapping auditory stimuli compete for limited perceptual resources and degrade task performance [53]. **H3** draws on research highlighting the complementary role of tactile cues in multimodal systems, where haptics serve as effective secondary channels when primary sensory modalities are occupied. Prior work demonstrates tactile cues consistently outperform auditory ones in detection tasks due to their resistance to environmental noise, while remaining slower than visual cues due to humans' innate visual processing advantages [6, 30, 39]. Finally, **H4** extends the predictions of MRT [57], which suggests that distributing cognitive processing across different sensory modalities reduces interference—a principle further supported by prior crossmodal research [12].

4.3 Participants

A priori power analysis was conducted using G*Power [13] to determine the minimum required sample size for our study. Assuming a medium effect size ($f = 0.25$) in Cohen's f [10], an alpha level of 0.05, and a desired power of 0.80, the analysis indicated that a minimum of 15 participants would be necessary.

To ensure sufficient power, we recruited a total of 36 participants (12 males, 24 females; M_{age} = 24.4, SD = 5.3) from our university via social media platforms. Prior to the study, participants were screened for color blindness, hearing impairments, and tactile sensitivity issues; no exclusions were necessary based on these criteria. Participants received monetary compensation of $15 CAD for their participation. The study was approved by the University of Calgary Research Ethics Board (Protocol ID: 24-0831) and conducted in accordance with institutional ethical guidelines.

4.4 Procedure

Upon arrival, participants provided informed consent. Afterward, participants underwent an overall orientation session. In the session, the goals of the experiment, the structure of the primary and secondary tasks, and the specific modality conditions (visual, auditory, tactile). Participants then completed a practice session lasting approximately 3 to 5 minutes. This practice session helped participants become familiar with each modality and gain hands-on experience interacting with virtual directional boxes and using the Xbox controllers in AR. After the practice, the main session started.

In the main session, participants experienced all nine experimental conditions in the counterbalanced order using a Latin Square method. In each experimental trial, participants performed a modality-specific primary pattern recognition task while periodically receiving directional notifications. Each trial lasted approximately 2 minutes, during which participants responded to around five directional notifications per condition. The 2-minute duration represented continuous, uninterrupted task/notification presentation time. The total trial duration including the user interaction varied between users, taking approximately 5 minutes. Following each condition, participants completed a brief survey asking about ease of use, preference, and task load. Between conditions, participants took a brief one or two minute rest to minimize fatigue and maintain task engagement. Upon completion of all conditions, participants provided additional feedback through a final open-ended survey regarding their overall experiences and perceptions of the notification interactions. The total duration of the experiment was approximately 70 to 90 minutes per participant.

4.5 Measurements

4.5.1 Subjective Measures

- **Ease of Use**: Participants rated how easily they recognized and responded to directional notifications for each modality condition through a single-item question, "Overall, how easy or difficult was the task to complete with this notification modality?", on a 7-point Likert scale (1: "very difficult" to 7: "very easy"). This straightforward measure identifies the practical usability of each notification type from the user's perspective, highlighting intuitive or problematic interaction modalities [44].
- **Preference**: Participants also indicated their preference for each notification modality combination through a single-item question, "How much do you prefer this notification modality?", on a 7-point Likert scale (1: "Strongly prefer other methods" to 7: "Strongly prefer this method"). This measure captures subjective preferences, helping to determine which modality combinations participants found most comfortable, intuitive, and desirable in multitasking AR contexts.
- **Task Load**: Participants rated their perceived cognitive and physical workload for each condition. We measured the NASA Task Load Index (TLX) across six subdimensions—Mental Demand, Physical Demand, Temporal Demand, Performance, Effort, and Frustration. To maintain consistency across measures, we employed a 7-point Likert scale ranging from "very low" to "very high" [19]. Prior research has demonstrated that 7-point scales offer an optimal balance between reliability, validity, and user preference, with minimal psychometric gains from increasing the number of scale points beyond seven [11, 40]. Understanding perceived workload helps to identify how cognitively demanding each modality alignment condition is, providing insights into potential fatigue or cognitive overload scenarios.

4.5.2 Objective Measures

Time Measures Time measures reflect how quickly participants responded to directional notifications, revealing cognitive processing and task-switching efficiency.

- **Perception Time**: Defined as the interval between the moment a directional notification appeared and the moment the participant pressed a button to temporarily pause the primary task. This measurement reveals how fast participants detected/recognized notifications while engaged in a primary task, reflecting attention switching and perceptual processing speed [15, 16].
- **Reaction Time**: Defined as the interval between notification recognition (pausing the primary task) and completing the required action (selecting the correct virtual box indicated by the directional cue). This indicates the time efficiency of translating notification awareness into an actionable response, directly assessing the usability and responsiveness of the notification system in multitasking contexts [34].

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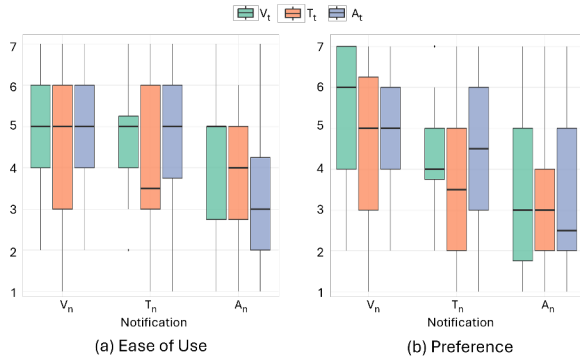


Figure 4: Boxplots of Ease of Use and Preference ratings.

Performance Measures These measures assess task accuracy as an indicator of cognitive processing across modality conditions.

- **Task Accuracy**: Performance during the modality-specific primary tasks (visual, auditory, tactile) was measured on a continuous scale from 0 to 1, representing the proportion of correctly identified patterns out of 20 trials (0: all incorrect, 1: all correct).
- **Task Miss Rate**: The proportion of missed responses on the same 0–1 scale—instances where participants failed to respond. This measure helps assess the impact of notification interruptions on cognitive performance, highlighting the extent to which such interruptions disrupted task engagement across different sensory modalities.
- **Notification Accuracy**: The proportion of correctly identified virtual boxes in response to spatially directional notifications was measured on a 0–1 scale, based on five trials per condition. This metric reflects how accurately participants interpreted and responded to directional notifications, providing insight into the clarity and effectiveness of each notification modality in AR.
- **Notification Miss Rate**: The proportion of missed notification—instances where participants failed to respond—was measured on the same 0–1 scale.

5 Results

A two-way repeated measures Aligned Rank Transform (ART) ANOVA [58], a non-parametric statistical method, was employed to analyze the results. This approach was selected due to the ordinal nature of the subjective Likert-scale responses and the non-normal distribution of the data, as verified by the Shapiro-Wilk normality test. The significance threshold was set at .05, and Bonferroni corrections were applied for multiple pairwise comparisons. The NASA-TLX results showed a strongly internal reliability, with a Cronbach's alpha of 0.81.

Details of the statistical results are presented below, and the significant effects are summarized in Table 1, which serves in place of inline significance reporting in the text.

5.1 Subjective Measures

5.1.1 Ease of Use

A significant interaction effect was observed between Notification Modality and Task Modality ($p = .034$). Additionally, a significant main effect of Notification Modality was observed ($p < .001$), whereas the main effect of Task Modality did not reach statistical significance ($p = .082$). The results are shown in Figure 4(a).

To further explore the interaction effect, simple effects analyses were conducted within each level of the factors. There were the simple effects of Task Modality within each Notification Modality (T_n: $F(2,70) = 3.98$, $p = .023$, A_n: $F(2,70) = 3.17$, $p = .048$). When the notification was tactile (T_n), the pairwise comparison revealed that participants rated ease of use significantly higher for A_t compared to T_t ($t(70) = -2.63$, $p = .028$). In the auditory notification condition (A_n), ease of use was significantly higher for V_t than for A_t ($t(70) = -2.47$, $p = .042$).

Conversely, when examining the effects of Notification Modality within each Task Modality, there were simple effects in V_t ($F(2,70) = 4.72$, $p = .012$), and A_t ($F(2,70) = 15.67$, $p < .001$). Participants rated V_n as significantly easier to use than A_n in the visual task condition (V_t; $t(70) = 2.88$, $p = .009$). Similarly, in the auditory task condition (A_t), both V_n ($t(70) = 5.38$, $p < .001$) and T_n ($t(70) = 4.03$, $p < .001$) were rated significantly higher in ease of use compared to A_n.

5.1.2 Preference

The interaction between Notification Modality and Task Modality was not statistically significant for participants' preference ratings ($p = .590$). Similarly, no significant main effect of Task Modality was observed ($p = .410$). In contrast, Notification Modality had a significant main effect on participant preferences ($p < .001$).

Post-hoc pairwise comparisons revealed significant differences between V_n and T_n ($t(280) = 4.11$, $p < .001$), V_n and A_n ($t(280) = 7.96$, $p < .001$), and T_n and A_n ($t(280) = 3.85$, $p < .001$). These findings suggest a clear preference ranking among the notification modalities: V_n was rated highest in preference ($M = 5.05$, $SD = 1.62$), followed by T_n ($M = 4.14$, $SD = 1.64$), and A_n received the lowest preference scores ($M = 3.16$, $SD = 1.74$), as illustrated in Figure 4(b).

5.1.3 Task Load

No significant interaction was found between the two factors in terms of overall task load ($p = .347$). Additionally, the main effect of the Task Modality on task load did not reach significance ($p = .080$). We found a significant main effect of Notification Modality on perceived workload ($p < .001$).

Pairwise comparisons showed statistically meaningful differences between V_n and T_n ($t(280) = -4.21$, $p < .001$), V_n and A_n ($t(280) = -8.44$, $p < .001$), and T_n and A_n ($t(280) = -4.23$, $p < .001$). These outcomes highlight a differentiated perception of task load across modalities. Participants reported the least mental demand for V_n ($M = 2.72$, $SD = 1.05$), a moderate level for T_n ($M = 3.17$, $SD = 0.96$), and the highest perceived load for A_n ($M = 3.70$, $SD = 1.15$).

The following section and Figure 5 present a breakdown of results across each NASA-TLX subscale:

Mental Demand A significant interaction was found between the two factors ($p = .044$). In addition, there was a significant main effect of Notification Modality ($p < .001$), while the main effect of Task Modality did not reach significance ($p = .058$).

To better interpret the interaction effect, simple effects analyses were conducted within each level of the factors. Under the tactile notification (T_n), the Task Modality had a significant effect on perceived mental demand ($F(2,70) = 7.31$, $p = .001$). Pairwise comparisons revealed that participants reported significantly lower mental demand in V_t than in T_t ($t(70) = -3.81$, $p < .001$).

In terms of Notification Modality within each Task Modality, all three task types showed significant differences (V_t: $F(2,70) = 7.17$, $p = .001$, T_t: $F(2,70) = 8.71$, $p < .001$, A_t: $F(2,70) = 14.84$, $p < .001$). Specifically, for V_t, participants reported significantly higher mental demand with A_n compared to both V_n ($t(70) = -3.30$, $p = .004$) and T_n ($t(70) = -3.25$, $p = .005$). Similarly, for the haptic task (T_t), mental demand was also significantly higher with A_n than with V_n ($t(70) = -3.75$, $p = .001$) and T_n ($t(70) = -3.47$, $p = .003$). This trend was further confirmed in the auditory task (A_t), where participants reported greater mental demand with A_n than with V_n ($t(70) = -5.42$, $p < .001$) and T_n ($t(70) = -3.19$, $p = .006$).

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Table 1: Summary of statistically significant results of ART ANOVA for subjective and objective measures.

Measures	Independent Variables	F-value	p-value	Effect Size (η_p^2)	Significant Effects (Main & Simple)
Ease of Use	Notification	$F(2,280) = 19.062$	$p < .001$	0.216	$V_n > T_n > A_n$
	Notification×Task	$F(4,280) = 2.643$	$p = .034$	0.036	Task Effects: ($T_nT_t > T_nA_t$); ($A_nV_t > A_nA_t$)
					Notification Effects: ($V_nV_t > A_nV_t$); ($V_nA_t > A_nA_t$), ($T_nA_t > A_nA_t$)
Preference	Notification	$F(2,280) = 31.672$	$p < .001$	0.184	$V_n > T_n > A_n$
Task Load	Notification	$F(2,280) = 35.602$	$p < .001$	0.203	$A_n > T_n > V_n$
- Mental Demand	Notification	$F(2,280) = 22.338$	$p = .001$	0.228	$A_n > T_n > V_n$
	Notification×Task	$F(4,280) = 2.49$	$p = .044$	0.034	Task Effects: ($T_nT_t > T_nV_t$)
					Notification Effects: ($A_nV_t > V_nV_t$), ($A_nV_t > T_nV_t$);
					($T_nT_t > V_nT_t$), ($A_nT_t > V_nT_t$); ($A_nA_t > V_nA_t$), ($A_nA_t > T_nA_t$)
- Physical Demand	Notification	$F(2,280) = 15.162$	$p < .001$	0.098	$A_n > T_n > V_n$
- Temporal Demand	Notification	$F(2,280) = 19.965$	$p < .001$	0.125	$A_n > T_n > V_n$
- Performance	Notification	$F(2,280) = 20.507$	$p < .001$	0.128	$V_n > T_n > A_n$
- Effort	Notification	$F(2,280) = 19.058$	$p < .001$	0.120	$A_n > T_n, V_n$
- Frustration	Notification	$F(2,280) = 34.173$	$p < .001$	0.196	$A_n > T_n > V_n$
Perception Time	Notification	$F(2,264) = 7.266$	$p = .001$	0.052	$A_n > T_n, V_n$
Reaction Time	Notification	$F(2,256) = 7.880$	$p < .001$	0.058	$A_n > T_n, V_n$
Task Accuracy	Task	$F(2,280) = 3.141$	$p = .045$	0.022	$T_t > A_t$
Task Miss Rate	Task	$F(2,280) = 18.505$	$p < .001$	0.117	$V_t > A_t > T_t$
Notification Accuracy	Notification	$F(2,280) = 63.622$	$p < .001$	0.312	$V_n > T_n > A_n$
Notification Miss Rate	Notification	$F(2,280) = 4.899$	$p = .008$	0.034	$V_n > A_n$

Physical Demand There was no significant interaction effect ($p = .868$), nor was there a significant main effect of Task Modality ($p = .340$). However, Notification Modality had a significant influence on perceived physical demand ($p < .001$). Participants reported significantly lower physical effort in the V_n condition ($M = 2.57$, $SD = 1.38$) compared to both T_n ($M = 2.90$, $SD = 1.24$; $t(280) = -3.32$, $p = .003$) and A_n ($M = 3.24$, $SD = 1.60$; $t(280) = -5.46$, $p < .001$).

Temporal Demand No significant interaction effect was found between the two factors ($p = .820$), and the main effect of Task Modality was also not significant ($p = .065$). By contrast, a significant main effect of Notification Modality emerged ($p < .001$). Participants experienced the least temporal pressure with V_n ($M = 2.85$, $SD = 1.42$), followed by T_n ($M = 3.44$, $SD = 1.54$; $t(280) = -3.66$, $p < .001$), and the highest demand was reported for A_n ($M = 3.93$, $SD = 1.60$; $t(280) = -6.29$, $p < .001$). Additionally, T_n was rated as significantly less temporally demanding than A_n ($t(280) = -2.64$, $p = .024$).

Performance The analysis revealed no significant interaction effect ($p = .136$) and no significant main effect of Task Modality ($p = .154$). However, the Notification Modality significantly affected the self-assessed performance of the participants ($p < .001$). V_n was associated with the highest performance ratings ($M = 5.39$, $SD = 1.22$), which were significantly higher than both T_n ($M = 4.93$, $SD = 1.29$; $t(280) = 3.88$, $p < .001$) and A_n ($M = 4.52$, $SD = 1.31$; $t(280) = 6.35$, $p < .001$). Participants also rated their performance with T_n as significantly better than with A_n ($t(280) = 2.47$, $p = .038$).

Effort No significant interaction effect was observed ($p = .269$), and the main effect of Task Modality was marginal and did not reach statistical significance ($p = .052$). However, Notification Modality had a significant effect on perceived effort ($p < .001$). Participants reported exerting the most effort in A_n ($M = 4.60$, $SD = 1.47$), which was significantly higher than in T_n ($M = 3.98$, $SD = 1.35$; $t(280) = -3.93$, $p < .000$) and V_n ($M = 3.62$, $SD = 1.60$; $t(280) = -6.09$, $p < .001$).

Frustration We found no significant interaction effect ($p = .081$), and no main effect of Task Modality ($p = .234$). In contrast, a significant main effect of Notification Modality was found ($p < .001$). Participants reported the lowest levels of frustration with V_n ($M = 2.37$, $SD = 1.35$), followed by T_n ($M = 2.77$, $SD = 1.48$; $t(280) = -3.75$, $p < .001$), and the highest levels with A_n ($M = 3.50$, $SD = 1.68$; $t(280) = -8.26$, $p < .001$). Additionaly, T_n also led to significantly lower frustration compared to A_n ($t(280) = -4.51$, $p < .001$).

5.2 Objective Measures

This section presents the results on objective measures (Figure 6).

5.2.1 Time Measures

Perception Time The analysis revealed no significant interaction ($p = .764$), and no significant main effect of Task Modality was found ($p = .700$). However, Notification Modality had a significant main effect on perception time ($p < .001$). The results showed V_n ($M = 1.49$, $SD = 1.52$) was significantly faster compared to A_n ($M = 1.87$, $SD = 1.50$; $t(264) = -3.53$, $p = .001$). In addition, T_n ($M = 1.61$, $SD = 1.63$) also yielded significantly quicker perception than A_n ($t(264) = -3.01$, $p = .008$).

Reaction Time No interaction effect ($p = .691$), and no main effect of Task Modality ($p = .787$) were found. However, there was a significant main effect of Notification Modality on reaction time ($p < .001$). Participants responded more quickly with V_n ($M = 5.82$, $SD = 2.40$) compared to A_n ($M = 6.66$, $SD = 2.56$; $t(256) = -3.44$, $p = .002$). Responses were faster with T_n ($M = 5.95$, $SD = 3.09$) than A_n ($t(256) = -3.43$, $p = .002$).

5.2.2 Performance Measures

Task Accuracy There was no significant interaction effect ($p = .123$) and no main effect of Notification Modality ($p = .171$). A significant main effect of Task Modality was observed ($p = .045$). Post-hoc comparisons showed a marginal difference in accuracy, with participants performing slightly better with T_t ($M = 0.887$, $SD = 0.149$) than A_t ($M = 0.864$, $SD = 0.149$; $t(280) = 2.36$, $p = .050$). No other significant differences were found.

Task Miss Rate No interaction effect was observed ($p = .752$), and Notification Modality did not yield a significant main effect ($p = .101$). Task Modality had a significant main effect ($p < .001$). Participants missed significantly more responses during V_t ($M = 0.096$, $SD = 0.138$) than in T_t ($M = 0.059$, $SD = 0.116$; $t(280) = 4.47$, $p < .001$), and also more in A_t ($M = 0.082$, $SD = 0.112$) than T_t ($t(280) = -5.81$, $p < .001$).

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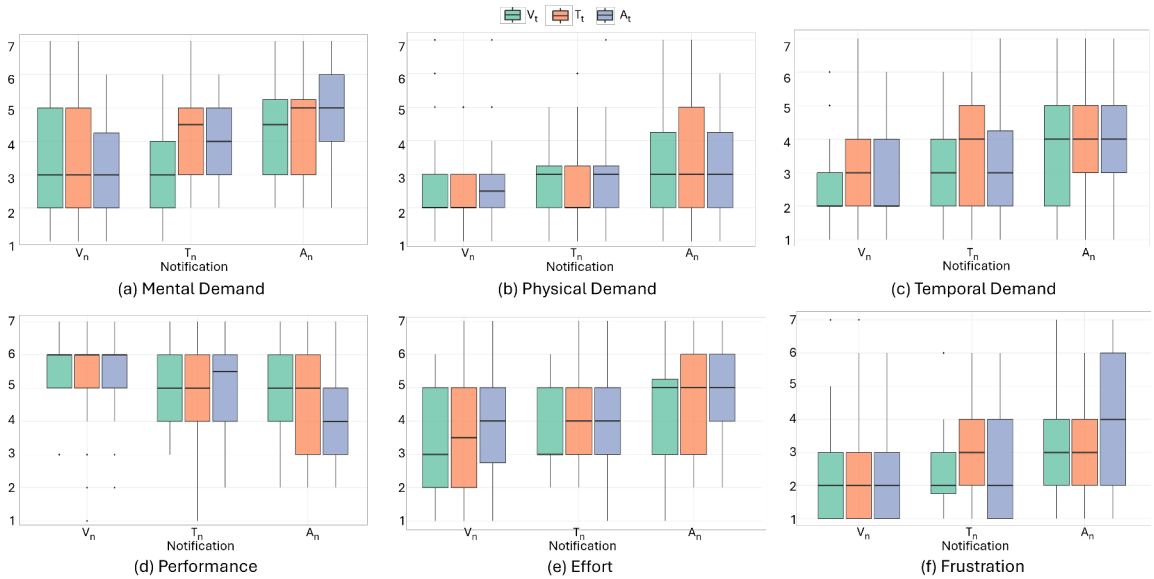


Figure 5: Boxplots of Task Load subscales (NASA-TLX).

Notification Accuracy No significant interaction effect ($p = .106$), and no main effect of Task Modality ($p = .832$) were observed. However, there was a main effect of Notification Modality in notification accuracy ($p < .001$). Participants achieved the highest accuracy with V_n condition ($M = 0.948$, $SD = 0.100$), clearly outperforming both T_n ($M = 0.865$, $SD = 0.227$; $t(280) = 2.82$, $p = .014$) and A_n ($M = 0.635$, $SD = 0.293$; $t(280) = 10.87$, $p < .001$). T_n was also significantly higher than A_n ($t(280) = 8.05$, $p < .001$).

Notification Miss Rate The analysis did not reveal a significant interaction ($p = .135$), nor a main effect of Task Modality ($p = .293$). Notification Modality significantly influenced the notification miss rates ($p = .008$). Participants missed more notifications with V_n ($M = 0.032$, $SD = 0.078$) than A_n ($M = 0.030$, $SD = 0.119$; $t(280) = 3.01$, $p = .008$). No other significances were observed.

6 Discussion

This section discusses our empirical findings, emphasizing how the interaction between notification and task modalities shaped user experience, performance, and perception in AR.

Visual Notifications Are Most Preferred and Performant (RQ1, H1) Visual notifications were identified as highly effective, providing rapid perception and response times while facilitating accurate spatial localization. Participants rated visual notifications as highest in ease of use, lowest in mental demand, and highest in perceived performance (**H1b**). Visual notifications also received the highest overall preference scores among all sensory modalities (**H1b**). Objective measures supported these findings, demonstrating superior accuracy, fastest perception ($M = 1.49\,s$), and quickest reaction times ($M = 5.82\,s$) (**H1a**).

Visual notifications, however, exhibited increased missed rates for both primary tasks and notifications. We attribute this to the limited FOV of HoloLens 2, which likely led to lapses in peripheral attention. Participants noted difficulties when notifications appeared outside their central focus, commenting: *"When a visual notification appeared slightly outside the center of my FOV, I often missed it as I was concentrating intensely on the primary task."* These results align with prior studies emphasizing the central-processing dominance of visual information [7, 43]. Consequently, these findings suggest AR interface designers should prioritize central positioning of visual cues or incorporate complementary sensory modalities such as haptic feedback to enhance notification effectiveness.

Auditory Notifications Impose High Cognitive Load (RQ1, H2) Auditory notifications demonstrated reliability regarding low missed notification rates; however, they showed significant limitations in directional accuracy and longer perception ($M = 1.87\,s$) and reaction times ($M = 6.66\,s$), indicating reduced task performance (**H2a**). Subjective measures corroborated these limitations, with auditory notifications receiving the lowest ratings in ease of use and the highest scores in mental, physical, and temporal demand, as well as frustration (**H2b**). A couple of participants explicitly reported feeling rushed to locate auditory notifications.

Notably, intramodal interference significantly intensified these negative outcomes when auditory notifications coincided with auditory primary tasks. Our results showed severe interference when task and notification demands overlapped in both sensory and cognitive (temporal processing) channels, consistent with MRT [57]. Auditory notifications were uniquely impacted, unlike the visual–visual and tactile–tactile pairings. This likely stems from two factors: humans generally process visual stimuli more easily than auditory ones [39, 50], and tactile inputs can be spatially separated (e.g., hands vs. torso), aiding signal segregation [23]. In contrast, auditory information relies on a shared temporal stream, requiring sequential attention [46], which makes it more vulnerable to interference. One participant explained: *"When task-related sounds and auditory notifications overlapped, it was very stressful and confusing to distinguish them clearly."* The results confirm prior research indicating auditory channels' temporal processing nature exacerbates cognitive overload when processing multiple auditory streams simultaneously [53]. Therefore, it is recommended that auditory notifications in AR should be combined with spatial information delivered through complementary modalities to minimize cognitive burden and enhance usability.

Tactile Notifications Offer Balanced and Complementary Support (RQ1, H3) Tactile notifications consistently demonstrated intermediate performance across all conditions. Objective measures showed that perception and reaction times were faster than for auditory cues but slower than for visual cues, and directional accuracy followed the same pattern. Missed notification rates were also lower than for auditory notifications (**H3a**).

Subjective ratings similarly placed tactile feedback between the other two modalities: participants rated tactile notifications as easier to use and less mentally demanding than auditory notifications,

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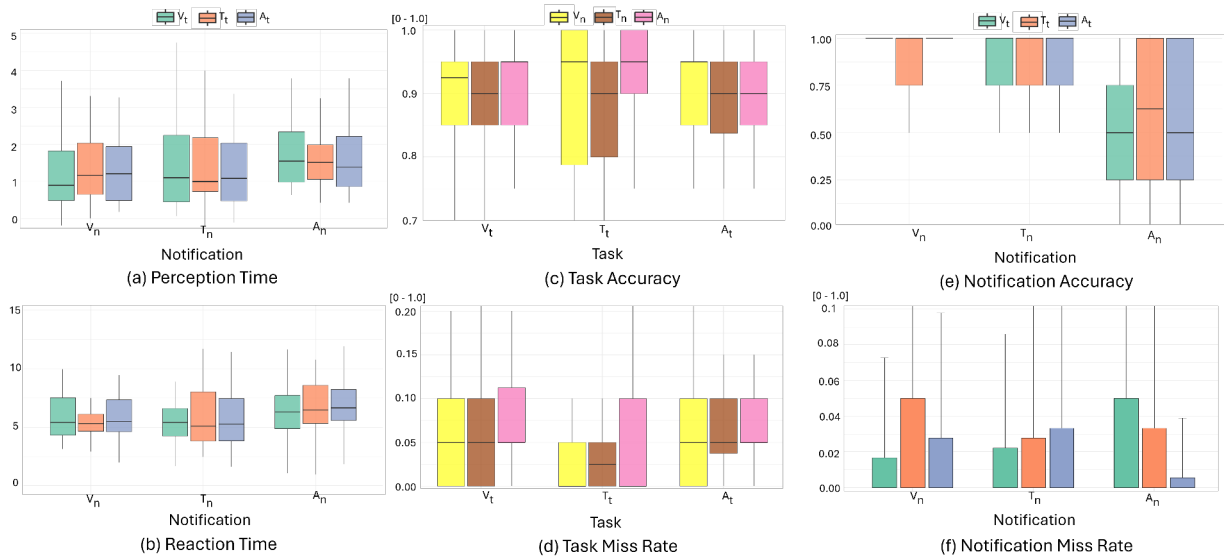


Figure 6: Objective measure results: (a) Perception Time, (b) Reaction Time, (c) Task Accuracy, (d) Task Miss Rate, (e) Notification Accuracy, and (f) Notification Miss Rate.

but not as favorable as visual ones. One participant remarked, *"Tactile notifications allowed me to quickly and clearly recognize directional cues,"* reflecting their perception of clarity and ease of use (**H3b**). These results position tactile feedback as a perceptually effective modality that balances responsiveness and cognitive comfort, making it well-suited for supporting directional notifications in multitasking AR environments. However, in the tactile-tactile condition, one participant remarked, *"I wasn't sure if the vibration meant the direction or was part of the task,"* reflecting the challenge of interpreting concurrent haptic signals.

Crossmodal Benefits and Conflicts (RQ2, H4) Our results revealed that modality mismatch between task and notification can provide measurable benefits, but also demonstrated important boundary conditions. Specifically, mismatched pairings were effective in reducing cognitive load across several conditions, supporting (**H4b**). In contrast, performance benefits such as improved notification accuracy were more limited and dependent on the specific modality pairing and perceptual clarity, partially supporting **H4a**.

Participants reported lower mental effort in mismatched combinations, such as visual notifications during auditory tasks, where sensory separation appeared to ease attentional switching. Some participants noted, *"When the cue was from another sense, it didn't interfere,"* highlighting the perceived benefit of modality mismatch. This pattern was particularly evident in tactile-notification conditions, where pairing tactile cues with non-tactile tasks (e.g., T_nA_t, T_nV_t) resulted in lower mental demand than tactile-tactile combinations. This suggests that even less spatially dominant modalities like tactile feedback can benefit from crossmodal separation when used in multitasking AR contexts (**H4b**).

Our finding is consistent with prior research showing that modality separation reduces interference and improves attentional control [49, 57]. However, our use of spatially directed notifications in AR —where users had to actively shift attention or respond to spatial cues—imposed greater demands on both perception and action than passive, non-interactive alerts. Mismatch did not always lead to better performance. Notification accuracy was highest in the matched visual-visual condition, suggesting that modality congruence may still support performance when the shared channel offers high spatial precision. This suggests that AR directional notification systems should strategically leverage crossmodal pairings to reduce cognitive load, while also ensuring perceptual clarity by aligning notification modalities with the spatial strengths of the task context.

6.1 Limitations and Future Work

This study is limited by inherent differences in sensory modalities. Although we made efforts to match task difficulty across visual, auditory, and tactile conditions, each modality has unique perceptual and cognitive properties. The lack of well-defined criteria for calibrating equivalent perceptual salience or intensity across modalities makes it challenging to ensure a balanced sensory load in multisensory comparisons. Additionally, individual differences in sensory acuity, attention, and spatial audio perception–shaped by head movement and ear structure–may have further influenced participants' performance across modalities [5].

Future work should explore more dynamic and realistic scenarios, such as walking, hands-busy industrial tasks, or collaborative MR settings. Long-term haptic vest use raised comfort issues—future designs should explore lightweight actuators or pulsed feedback. Additionally, adaptive notification systems that adjust modality based on user state or context (e.g., workload, ambient noise) could further improve notification relevance and reduce disruption.

7 Conclusions

This paper systematically examines three sensory modalities in AR for delivering notifications during modality-specific tasks. The results showed the visual notification yielded the fastest response time and highest accuracy. However, their effectiveness was limited by the narrow FOV of the AR device during visually demanding tasks. Audio notifications increase cognitive load and direction ambiguity, particularly in auditory tasks. Haptic notifications provide a balanced alternative with minimal task disruption, although wearability concerns remain. While our results showed that the visual–visual condition for notification and task achieved the highest accuracy, findings from the tactile and auditory modalities suggest that using a different modality from the one employed in the main task may be more effective. Moreover, our findings highlighted the importance of adaptive modality selection strategies that take into account both the urgency of notifications and individual user preferences. Future AR systems should dynamically tailor notification modalities to minimize interference and optimize user experience across diverse usage scenarios.

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